

CCNA 200-301 V2.0

26 IPsec VPNs

Part IV — Network Services and Security

EXAM TOPICS

4.5

LECTURER

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Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

What you will be able to do

- **Distinguish** site-to-site from remote-access VPNs and say which suits a given requirement.
- **Name** the protocols in the IPsec suite — IKE, ESP and AH — and state what each contributes.
- **Compare** tunnel mode and transport mode, and say which is used where.
- **Explain** why AH is effectively unusable through NAT, and what is used instead.

Before we start

Chapter 6 for ports and protocol numbers. Chapter 22 — the interaction between IPsec and NAT is most of what makes this topic interesting. Chapter 23 for the idea of authentication as distinct from encryption.

Vocabulary for this chapter

IPsec

IKE

ISAKMP

ESP

AH

tunnel mode

transport mode

security association

NAT-T

site-to-site

remote access

GRE over IPsec

Each is defined where it first matters — not here.

The only describe topic in this part

This is what the blueprint actually asks for

Topic **4.5** is *describe IPsec remote access and site-to-site VPNs (protocols and transport modes)*. There is no configuration verb, and no crypto map or IKEv2 proposal syntax is required of you.

That is why this chapter is short and has no lab: writing one would teach you something the exam does not ask for, at the cost of time you need elsewhere in this part. What it *does* ask for is in the parenthesis — **the protocols and the transport modes** — so those two lists are what this chapter drills.

If you go on to CCNP Security, this chapter is the vocabulary you will build on. For now, know the names, the jobs and the trade-offs.

The four services IPsec provides

- **Confidentiality** — encryption, so the payload cannot be read. Normally AES.
- **Integrity** — a keyed hash, so alteration is detectable. Normally SHA-2.
- **Authentication** — proof that the peer is who it claims to be. A pre-shared key or a digital certificate.
- **Anti-replay** — sequence numbers, so a captured packet cannot be re-sent later to useful effect.

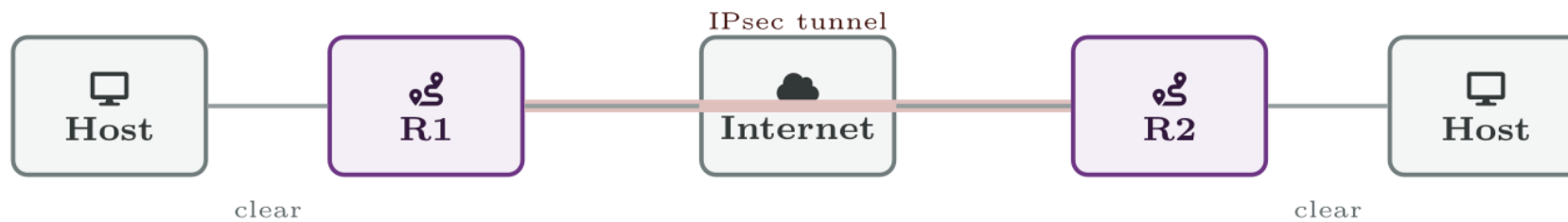
Underpinning all four is **key management**: Diffie–Hellman, which lets two peers agree a shared secret over a channel an eavesdropper is watching.

Two deployment models

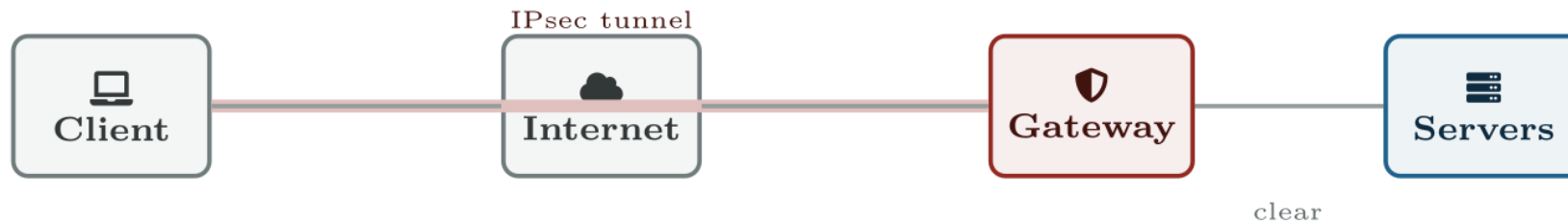


Two deployment models

Site-to-site



Remote access



Where the tunnel starts decides the model. Site-to-site protects gateway to gateway; remote access protects all the way to the user's device.

Traffic is only protected inside the tunnel

In the site-to-site diagram, traffic between the host and its own gateway is in clear. IPsec protects the internet crossing, not the LAN at either end. That is usually the right trade — but it means a VPN is not an answer to a threat already inside the building.

The protocols



Two facts that are asked directly

ESP encrypts; AH does not. If a question asks which protocol provides confidentiality, the answer is ESP, every time.

ESP and AH are not TCP or UDP. They are IP protocols in their own right, numbered 50 and 51, sitting at the same layer as TCP (6) and UDP (17). This is why a firewall rule that permits “UDP 500” alone will let the tunnel negotiate and then carry no data — a memorable and very common real fault.

Security association

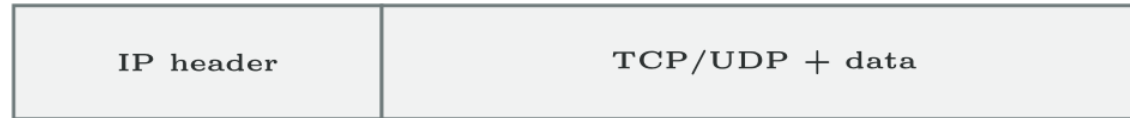
A one-directional agreement covering the algorithms, keys and lifetimes for one flow. Because it is one-directional, a working tunnel has *two* — one in each direction. Asymmetric failures, where traffic passes one way only, are usually a phase 2 mismatch.

Transport mode and tunnel mode



Transport mode and tunnel mode

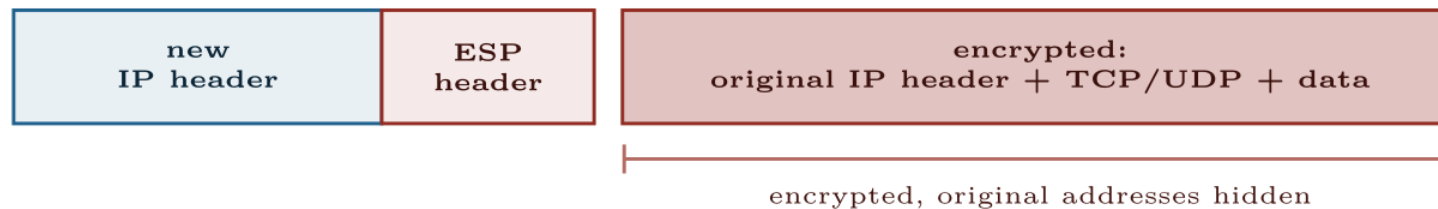
Original packet



Transport mode



Tunnel mode



Transport mode keeps the original IP header. Tunnel mode encrypts it and adds a new one, which is what lets private addresses cross a public network.

If the question does not say otherwise, it is tunnel mode

This is what the blueprint actually asks for

Tunnel mode is the default for the VPNs this topic names. The single sentence to remember: **tunnel mode encrypts the original IP header and adds a new one; transport mode does not.** Every other difference in the table follows from that.

IPsec and NAT



Why AH is dead

AH authenticates the *entire* packet, including the source and destination addresses in the IP header. NAT's whole job is to rewrite those addresses. So a NAT device on the path invalidates AH's integrity check by definition, and the far end discards the packet.

There is no configuration that fixes this; the two mechanisms are logically incompatible. Since almost every internet path involves NAT somewhere, AH has been left behind, and ESP — which does not protect the outer header — is what everyone uses.

GRE over IPsec, in one paragraph

IPsec tunnel mode carries unicast IP and nothing else — no multicast, and therefore no routing protocols. A site-to-site VPN that needs OSPF between the sites wraps the traffic in a GRE tunnel first, which *can* carry multicast, and then protects the GRE with IPsec.

You will meet the phrase “GRE over IPsec”. For this exam, know why it exists: GRE brings the ability to run a routing protocol, IPsec brings the security, and neither does both.

When things do not work



Common pitfalls

- **Saying AH encrypts.** It does not. Only ESP does.
- **Calling ESP and AH “TCP” or “UDP”.** They are IP protocols 50 and 51.
- **Mixing up the modes.** Tunnel mode adds a new IP header; transport mode keeps the original.
- **Assuming a VPN protects the whole path.** It protects the tunnel.
- **Expecting a routing protocol to run over plain IPsec.** It cannot — multicast needs GRE.
- **Treating a successful ping as a working tunnel.** Test with something large enough to expose an MTU problem.

The tunnel is up and shows both peers established. No user traffic crosses it.

A site-to-site VPN between R1 and R2. Both routers report an established IKE session with the other. Users at either site cannot reach the other site at all.

The tunnel is up and shows both peers established. No user traffic crosses it.

What would you check first — and what do you expect to see?

show crypto isakmp sa

R1#

IPv4 Crypto ISAKMP SA

dst	src	state	conn-id	status
198.51.100.9	203.0.113.2	QM_IDLE	1004	ACTIVE

show crypto ipsec sa | include pkts|local ident|remote ident

R1#

```
local ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.20.0/255.255.255.0/0/0)
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Read the two outputs as a pair, because together they locate the fault precisely.

`QM_IDLE` and `ACTIVE` mean **phase 1 succeeded**: the peers found each other, agreed on IKE parameters and authenticated. So the pre-shared key is right, UDP 500 is reaching both ends, and the peer addresses are correct. All of that is now ruled out.

But every counter is zero — in *both* directions. Nothing has been encrypted and nothing decrypted. Traffic is not being selected for the tunnel, or is being selected and then dropped before it is sent.

Why it is doing that

The identities are the place to look. **R1** intends to protect **192.168.10.0/24** to **192.168.20.0/24**. If **R2**'s definition is not the exact mirror — a different mask, a wider or narrower range — phase 2 cannot agree on what the tunnel is for, and no traffic qualifies.

The class of fault, which is the point here. Phase 1 up with phase 2 idle is always a *policy* disagreement rather than a connectivity or credential one: the crypto ACL, the transform set, or the mode. Knowing which half of IKE succeeded halves the search before you have looked at anything.

Why it is doing that

The other candidate, if the identities do match, is a firewall permitting UDP 500 — which is why phase 1 worked — while dropping IP protocol 50. Symptoms are identical; the discriminator is whether `#pkts encaps` is zero (traffic never selected) or climbing while `#pkts decaps` stays at zero (sent, never arriving).

CHECKPOINT 1 of 6

Which IPsec protocol provides confidentiality, and what is its IP protocol number?

Discuss · then advance

Checkpoint 1 of 6

Which IPsec protocol provides confidentiality, and what is its IP protocol number?

ESP, IP protocol **50**. AH provides integrity and authentication but no encryption.

CHECKPOINT 2 of 6

State the difference between tunnel mode and transport mode in one sentence, and say which a site-to-site VPN uses.

Discuss · then advance

Checkpoint 2 of 6

State the difference between tunnel mode and transport mode in one sentence, and say which a site-to-site VPN uses.

Tunnel mode encrypts the entire original packet and adds a new IP header; transport mode keeps the original header and encrypts only the payload. A site-to-site VPN uses **tunnel mode**.

CHECKPOINT 3 of 6

Why can AH not survive a NAT device on the path?

Discuss · then advance

Checkpoint 3 of 6

Why can AH not survive a NAT device on the path?

Because AH authenticates the entire packet including the source and destination addresses in the IP header, and NAT's whole function is to rewrite them. The integrity check therefore fails by definition and the far end discards the packet. No configuration fixes this.

CHECKPOINT 4 of 6

What does phase 1 of IKE accomplish, and what does phase 2 accomplish?

Discuss · then advance

Checkpoint 4 of 6

What does phase 1 of IKE accomplish, and what does phase 2 accomplish?

Phase 1 authenticates the peers and builds a secure channel between them — the IKE security association. Phase 2 uses that channel to negotiate the security associations that protect user traffic, including which traffic is protected.

CHECKPOINT 5 of 6

Why is GRE needed if a site-to-site VPN must carry OSPF?

Discuss · then advance

Checkpoint 5 of 6

Why is GRE needed if a site-to-site VPN must carry OSPF?

Because plain IPsec tunnel mode carries unicast IP only, and OSPF depends on multicast. GRE can carry multicast, so the routing protocol runs inside a GRE tunnel, which IPsec then protects.

CHECKPOINT 6 of 6

A tunnel is established and small pings succeed while file transfers hang. What is the likely cause?

Discuss · then advance

Checkpoint 6 of 6

A tunnel is established and small pings succeed while file transfers hang. What is the likely cause?

Path MTU. The added headers push large packets past the MTU somewhere on the path, so small packets pass and large ones are dropped or fragmented badly. A good reminder that “it pings” is not “it works”.

What to take away

- IPsec provides confidentiality, integrity, authentication and anti-replay, with Diffie–Hellman for key agreement.
- Site-to-site joins two gateways and is invisible to hosts; remote access runs from a user's device to a gateway and authenticates the user.
- IKE (UDP 500, or 4500 with NAT-T) negotiates. ESP (IP protocol 50) protects data and **encrypts**. AH (IP protocol 51) authenticates but does **not** encrypt.
- Phase 1 builds the secure channel between peers; phase 2 builds the security associations that protect user traffic. SAs are one-directional, so a working tunnel has two.

What to take away (continued)

- Tunnel mode encrypts the whole original packet and adds a new IP header — the normal case. Transport mode keeps the original header.
- AH is incompatible with NAT because it authenticates the header NAT rewrites. ESP with NAT-T (UDP 4500) is the working combination.
- Plain IPsec carries no multicast, so a VPN needing a routing protocol uses GRE over IPsec.

Where we got to

- IPsec provides confidentiality, integrity, authentication and anti-replay, with Diffie–Hellman for key agreement.
- Site-to-site joins two gateways and is invisible to hosts; remote access runs from a user's device to a gateway and authenticates the user.
- IKE (UDP 500, or 4500 with NAT-T) negotiates. ESP (IP protocol 50) protects data and **encrypts**. AH (IP protocol 51) authenticates but does **not** encrypt.
- Phase 1 builds the secure channel between peers; phase 2 builds the security associations that protect user traffic. SAs are one-directional, so a working tunnel has two.

NEXT

Chapter 27 — Wireless Principles